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A Real-Time Rig Control ROP Optimization Framework Using Machine Learning and Predictive Vibration Modeling

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Abstract

This paper presents a novel real-time rig control framework for optimizing the rate of penetration (ROP) in horizontal drilling operations by integrating machine learning (ML), transfer learning, and predictive vibration modeling within a constraint-aware optimization loop. A general ROP prediction model is first trained on a large, multi-well dataset using ensemble learning methods, with XGBoost emerging as the best-performing algorithm. To improve prediction accuracy in new wells, a transfer learning strategy fine-tunes the model using real-time and offset well data, ensuring adaptability to local geological conditions. The fine-tuned model is embedded within an optimization algorithm that recommends drilling parameters to maximize ROP while respecting operational constraints. These include rotary speed limits derived from a predictive vibration model that identifies resonance zones to prevent harmful drilling vibrations. Additional constraints ensure parameter feasibility, operational continuity, and adherence to target depth of cut specifications. The framework is validated in both simulated and real-world environments. In simulation, a physics-based drilling model is used to generate ground-truth data for assessing convergence and performance. Results show that the optimization algorithm reliably steers drilling parameters toward the region of maximum ROP. Field validation on real rig data further confirms the robustness and practicality of the proposed approach, demonstrating significant improvements in ROP prediction accuracy and drilling performance. This hybrid data-physics optimization framework enables safer, more efficient drilling by providing actionable, real-time parameter recommendations that account for both data-driven insights and physical system dynamics.

Introduction

Optimizing the Rate of Penetration (ROP) in drilling operations is a critical objective that directly influences operational efficiency and overall cost. A higher ROP shortens the time required to reach the target depth, thereby reducing rig time and associated expenses. However, achieving maximum ROP is challenging due to the complex, non-linear, and multi-parameter nature of drilling processes, which are affected by a variety of factors such as lithology, weight on bit (WOB), rotary speed, mud properties, and bit type. Traditional ROP

prediction and optimization methods often rely on empirical correlations or physical models that may not generalize well across varying geological formations and operational conditions (Bingham 1965, Motahhari 2010). In recent years, machine learning (ML) techniques have emerged as powerful tools to address these challenges by capturing hidden patterns and complex relationships within large volumes of drilling data. These data-driven models can learn from historical and real-time datasets to predict ROP with higher accuracy and adaptability than conventional methods. Comprehensive reviews (Najjarpour 2022, Barbosa 2019, Zhong 2022) highlight that neural networks (ANNs) are the most commonly used ML algorithms in ROP prediction, favored for their ability to model complex, non-linear functions. Other algorithms such as regression, boosting, random forests (RF), and support vector machines (SVM) have also been implemented effectively (Amer 2017, Gandelman 2012, C. W. Hegde 2015). Typical input parameters for these models include rotary speed (RPM), WOB, formation properties, depth, flow rate, mud characteristics, torque, and standpipe pressure. Comparative studies demonstrate that ML-based ROP prediction models generally outperform traditional physics-based models, offering better predictive accuracy across diverse lithologies and operational settings (Soares 2019, Barbosa 2019, Liu 2023). Moreover, recent work has successfully deployed ML models for real-time ROP optimization in the field, enabling dynamic adjustment of drilling parameters and improved drilling performance (Al-Riyami 2023).

Beyond prediction, various optimization algorithms have been employed to enhance ROP by identifying optimal drilling parameters. These include metaheuristic methods such as particle swarm optimization (PSO) (Giftson Joy 2021), genetic algorithms (Elmgerbi 2021, Duru 2022), and other evolutionary techniques (C. D. Hegde 2018). Integrating ML-based predictive models with these optimization algorithms has shown significant promise, allowing continuous updating of model parameters and real-time optimization of controllable variables such as WOB, rotary speed, and flow rate (Carpenter 2023, Duru 2022). Nevertheless, existing ROP optimization methods still face challenges in real-time drilling applications. Traditional analytical models often lack the accuracy and adaptability required for new wells (Moazzeni 2020). To address these limitations, hybrid approaches combining ML with advanced optimization techniques have been developed. For example, (Wang 2022) proposed an online ROP optimization method integrating support vector regression with particle swarm optimization, while (Sui 2013) introduced a moving-horizon multiple regression approach combined with model predictive control for real-time optimization. (Elmgerbi 2021) further advanced the field by coupling predictive ML models with genetic algorithms to optimize ROP dynamically. These state-of-the-art methods represent significant steps toward enhancing drilling efficiency by leveraging the predictive power of machine learning and the robustness of optimization algorithms, enabling more accurate, adaptable, and real-time control of drilling parameters.

However, despite the demonstrated advantages of these data-driven methods, two major challenges remain. First, most existing approaches lack integration with physical principles and rely solely on data to optimize ROP, potentially overlooking essential dynamics of the drilling system. Second, optimization algorithms often do not incorporate operational constraints on the suggested drilling parameters. As a result, the recommended parameters, such as rotary speed or weight on bit, may be infeasible or even unsafe to implement in real-time field operations, as they fail to account for the actual physical conditions encountered during drilling. To address these limitations, this paper proposes a hybrid data-physics optimization framework that leverages both machine learning and physical modeling. The primary goal is to develop a ROP optimization algorithm for lateral wells that ensures predictive accuracy while adhering to realistic and safe drilling constraints. The proposed method uses historical and real-time data to train a ROP prediction model based on input variables such as surface rotation speed, downhole motor rotation speed, and surface-estimated weight on bit (WOB). To enhance the generalization and performance of the prediction model, a transfer learning approach is employed using offset wells. This allows knowledge learned from previously drilled wells to be effectively transferred and adapted to current drilling conditions, improving prediction accuracy and model robustness across varying geological contexts. To integrate physical constraints, the

optimization algorithm is augmented with a predictive vibration model. Specifically, at each depth where optimization is performed, the model calculates the resonance rotary speeds that must be avoided to prevent detrimental drilling vibrations. These resonance thresholds are then embedded as constraints within the optimization process, ensuring that the suggested rotary speeds remain within safe operational limits. This proposed constraint-based optimization enables not only more efficient drilling, but also safer and more reliable decision-making in real-world scenarios.

The remainder of this paper is organized as follows. The 2nd section presents a detailed description of the proposed ROP optimization workflow, outlining its key components: the training of the ROP prediction model, the definition and application of transfer learning, the development of the predictive vibration model, and the integration of the optimization algorithm. The 1st section validates the proposed methodology using a virtual rig environment to assess its performance under controlled conditions. Finally, the last section demonstrates the applicability of the approach on real-world data collected from an actual drilling rig, highlighting its effectiveness and robustness in field operations.

Methodology

Principle

The workflow of the proposed method, illustrated in Fig. 1, consists of different stages designed to optimize the rotary drilling ROP while respecting operational constraints and avoiding drilling dysfunctions. First, data is collected from offset wells that are as close as possible to the current well conditions in terms of formation, wellbore trajectory, bottom hole assembly (BHA), drill bit and mud motor specifications... These wells serve as a baseline to compare performance and at the same time as valuable data to model the relationship between the key drilling parameters such as surface rotation speed, downhole motor rotation speed, and WOB and the resulting ROP, mechanical specific energy (MSE), torque on bit (TOB), differential pressure (ΔP), and surface torque. Actual drilling data from the current well are collected via WITSML or directly from the rig. It is used to fine-tune the models and to guarantee an in-time reaction to potential issues or dysfunctions. The actual wellbore trajectory, drilling data, and BHA are used in a predictive mapping model to recommend surface rotation speed that avoids resonant vibrations. The outputs of this model, along with the actual and offset well data, feed an optimization algorithm that looks for an optimum in terms of an objective function that balances between maximizing ROP, minimizing MSE, keeping stable drilling conditions, and respecting constraints on all drilling parameters (like maximum ΔP or surface torque).

The optimization algorithm is built on a data-based ROP prediction model. This can be either an aggregation of the actual and offset well data or a machine learning (ML) model. In the first option, more recent data is given more weight compared to old data, and current data more weight than offset well data. This allows the optimization to be as much reactive as possible to the current conditions when suggesting recommendations. In the second option (ML modeling), the data is used to fine-tune the model with different levels: (i) a general model is trained on historical data including a large and diverse dataset collected from various geographical regions, including wells with different lateral trajectories, bit sizes, and bottom hole assembly (BHA) configurations; (ii) a transfer learning method is applied to adapt the model to the data coming from the offset wells; and (iii) a live retraining based on real-time data is applied to get as close as possible to current drilling conditions.

The time-based data (actual and offset) is collected every second then it goes through an operation detection algorithm to get only the rotary drilling data. It is then aggregated into depth-based data with a data acquisition step (1 ft by default) and filtered to remove outliers and transients. The quantity of data used for fine-tuning is parametrizable (100 ft worth of data for example). The recommendations are computed and sent to the rig every optimization step that is also configurable depending on the situation (example, every 5 or 10 ft). They are done for a projected distance ahead of the current drilling depth (100 ft for example),

which allows to have a better visibility for the driller for the upcoming changes. This offers also a buffer of recommendations in case of connection issues with the rig.

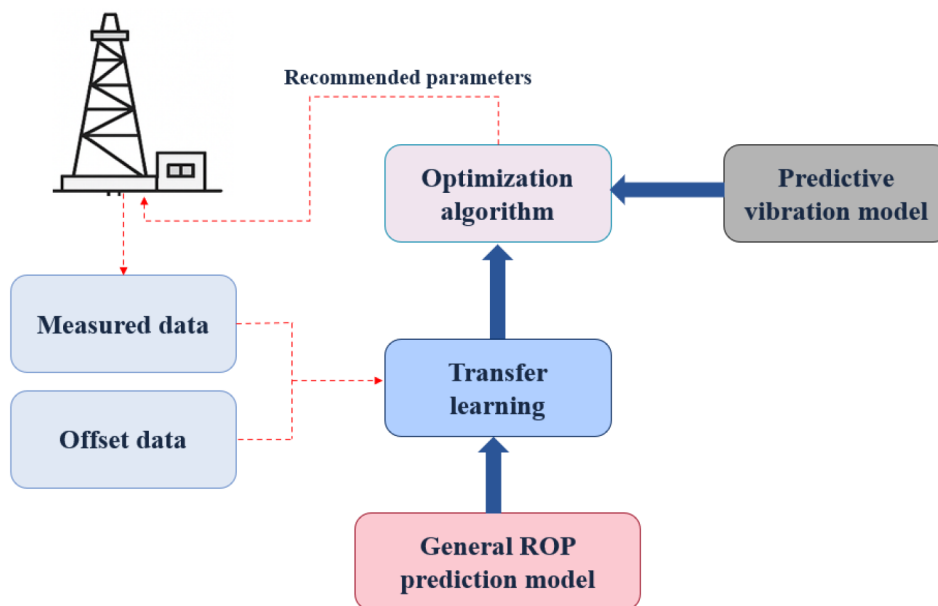


Figure 1—Workflow of the proposed ROP optimization method.

Predictive vibration model

While drilling, the drill string is subject to multiple external excitations that can lead to resonance if the surface rotation speed is not chosen carefully. To mitigate this problem, a physical model is used to compute the resonant (or natural) frequencies of the drill string in the axial, torsional, and lateral directions (see example in Fig. 2). These frequencies are expressed in terms of critical surface rotation speeds that should be avoided. As illustrated in Fig. 2, an algorithm is employed to automatically recommend a roadmap of surface rotation speed as a function of the drilled depth. It aims to stay away from all the critical speeds, keep the recommendation constant for the longest distance possible, and ensure a margin of safety around the recommended values. This roadmap is updated in real-time as the trajectory grows and as actual drilling data is gathered. In the same figure, the curves for the critical speeds and the recommendations are plotted with solid lines in the already drilled part of the well, and with dashed lines in the projected part of the well (starting from approximately 18200-ft measured depth). This projected part allows the driller to have a clear vision for the upcoming necessary changes required ahead of time and offers a resilience to eventual problems of connectivity to the rig.

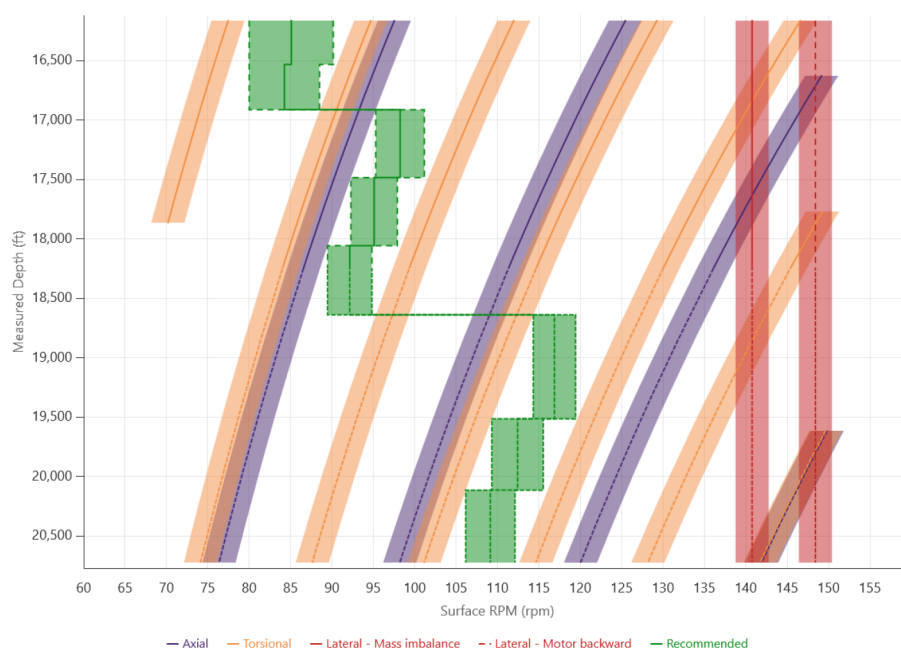


Figure 2—Predictive Vibration Mapping with the recommended surface rotation speed.

ROP prediction model

A regression-based machine learning model based on the Extreme Gradient Boosting (XGBoost) method is developed to predict the ROP using the actual bit depth, the bit depth at which the lateral section begins D_H , surface rotation speed, downhole motor rotation speed, and surface estimated WOB. The downhole motor rotation speed ω_{Motor} is derived from flow rate according to mud motor specification.

To train the ROP prediction model, we use a large, depth-based drilling dataset sampled at 1-ft intervals. The dataset comprises horizontal sections from 823 wells, as this study focuses exclusively on horizontal drilling performance, as outlined in the introduction. Only data corresponding to the horizontal portions of each well is included in the training set. These wells are drilled from 65 different rigs, covering a wide range of geographical regions and geological formations, which contributes to the diversity and robustness of the dataset. For each well, the data includes key surface measurements: surface rotation speed, estimated downhole motor rotation speed, estimated WOB, and the measured ROP. In total, the dataset consists of approximately 4.3 million rows of reliable, high-quality data suitable for model training. Fig. 3 presents heatmap showing the correlation between input features and the target variable, ROP. Among all features, surface WOB exhibits the strongest correlation with ROP. Additionally, the input features show minimal inter-correlation, indicating a low degree of redundancy among them. Fig. 4 illustrates the variation of predicted and true ROP over depth for a test well using the XGBoost model, showcasing the model's ability to capture the ROP trend.

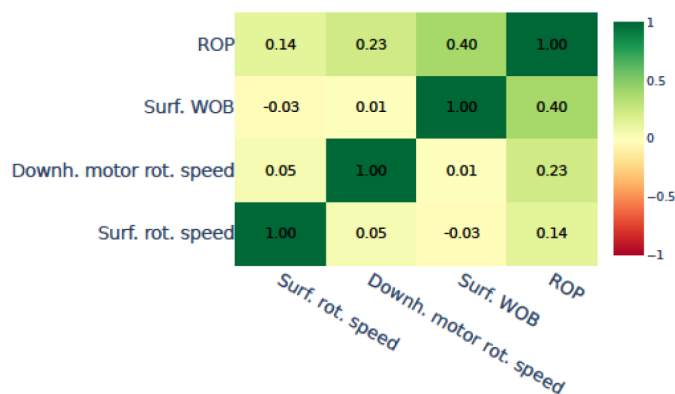


Figure 3—Heatmap illustrating the relationships between the input features and the target ROP across the entire dataset from all wells.

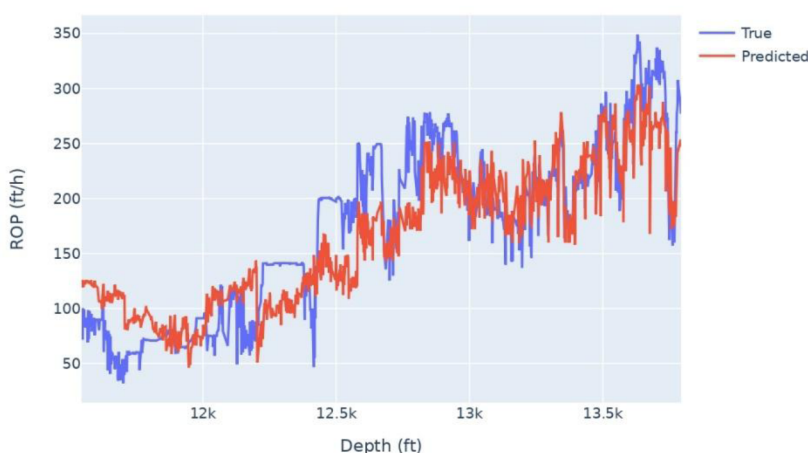


Figure 4—True and predicted ROP variations with depth for a test well using the XGBoost model.

Transfer learning

The first part of this work focuses on developing a generalizable ROP regression model that can be directly applied to new wells with different characteristics, such as new formations or different drill string compositions than the wells used while training, without needing to retrain the entire model from scratch. To further enhance performance for a specific new well, transfer learning (TL) can be employed by fine-tuning the pre-trained model using a small amount of drilling data from that well. Transfer learning leverages knowledge gained from a related domain (the source domain) to improve learning in a different but related domain (the target domain). This approach reduces the amount of labeled data required for the target task and accelerates training, as described by (Zhuang, et al. 2020). Essentially, a model trained on the source task serves as a starting point and is then re-trained, either partially or fully, on the target task of interest.

In our study, the source model is a pre-trained ROP regression model developed using a large dataset that includes drilling data from multiple wells. To adapt this model to a new target well, we employ a fine-tuning process that incorporates two types of data: surface measurements collected from the target well itself and, when available, historical data from offset wells. This approach allows the model to better capture the unique geological and operational characteristics of the new environment. During drilling, surface data are continuously recorded, providing both the input features for the ROP prediction and the corresponding ground-truth ROP values. These data are then used to adjust selected parameters of the pre-trained model, enabling it to refine the relationship between inputs and ROP in the specific context of the new well. The fine-tuning process preserves the broad knowledge learned from the original model, trained on diverse well conditions, while enhancing its accuracy for the current drilling scenario. Two fine-tuning strategies can be

employed: a continuous approach, where the model is updated with each new measurement, or a periodic approach, where updates occur at fixed intervals.

Fig. 5 presents the performance comparison of the XGBoost model before and after the application of TL. In this example, after applying TL every 100 ft, the model is fine-tuned using data from the target well, leading to a significant improvement in prediction accuracy. Specifically, the MAE is reduced by approximately 58%, indicating that the adapted model predicts the ROP more accurately than the original pre-trained version. In addition to this performance gain, TL is also computationally efficient, allowing for rapid adaptation to new wells without the need for full retraining.

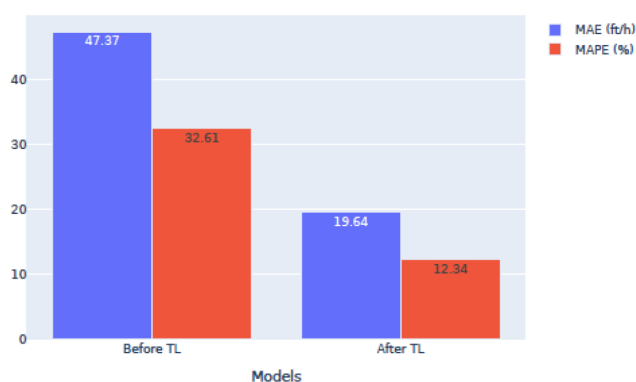


Figure 5—Comparison of MAE and MAPE for the XGBoost model before and after the application of transfer learning (TL) on the 100 test wells.

Optimization Algorithm

Once the ROP prediction model has been fine-tuned to the specific conditions of a new well, the next step is to use this model within an optimization framework. The goal is to identify the optimal set of drilling parameters that maximizes the objective function (as described above) at a specific target depth. More precisely, while drilling at a current depth D , the optimization algorithm is launched to determine the best drilling parameters at a future depth $D + \delta D$, where δD is a depth interval defined by the driller. The optimization algorithm takes as fixed inputs the depth where the horizontal section starts D_H , and the depth where to optimize $D + \delta D$. The remaining ROP prediction model input parameters, surface rotation speed, downhole motor rotation speed, and surface-estimated WOB, are treated as variables. These variables are iteratively adjusted by the algorithm to maximize the objective function.

For each optimization run, the algorithm performs N iterations to explore different combinations of the adjustable parameters. In each iteration, a new set of drilling parameters is generated and passed through the previously trained ROP prediction model (with transfer learning applied). Based on the predicted ROP, the algorithm evaluates whether this new combination yields better performance compared to previous iterations. It then updates its search accordingly to converge toward the optimal parameter set. To ensure practical feasibility and operational safety, constraints can be imposed on the optimization variables. Specifically, upper and lower bounds can be defined for the surface rotation speed, downhole motor rotation speed, and surface WOB. For example, in the case of surface rotation speed, the recommended range of surface rotation speed (see Fig. 2) is considered as constraints to the optimization algorithm. All generated parameter combinations must respect these constraints throughout the optimization process.

An additional constraint can be introduced to limit the variability between the current drilling parameters used at depth D , and the optimized parameters proposed for depth $D + \delta D$. This ensures that the proposed changes are not excessively different from what is currently in use, particularly important when the depth increment δD is small. This variability constraint is controlled by a maximum allowed deviation ratio, also specified by the driller. The optimization algorithm must respect this ratio when proposing new drilling parameters to maintain operational consistency and reduce abrupt changes in the drilling program.

Validation on a virtual rig

The process used to validate the optimization algorithm on a virtual rig is schematically represented in Fig. 6. Specifically, the data are generated by a physical model that emulates the behavior of a target well. This allows controlled testing of the optimization process in a realistic yet reproducible environment. The bit-rock interaction model is the one presented by (Nystad 2021) and based on the model of (Detournay 2008). The surface torque is computed based on a torque and drag model.

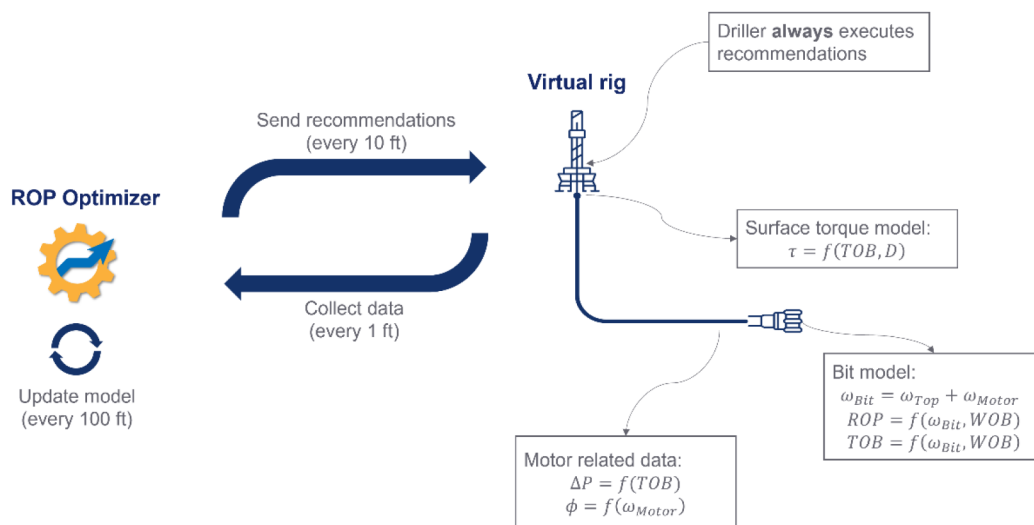


Figure 6—Schematic representation of the process used to validate the ROP optimization algorithm on a virtual rig.

Since the simulated data are generated with known ground truth, including the drilling parameters that yield the maximum ROP, we can directly assess whether the optimization algorithm converges toward the optimal values. At each iteration, the drilling parameters proposed by the algorithm are applied, allowing us to monitor the model's ability to adapt through transfer learning and to consistently improve both ROP prediction and parameter optimization. The evaluation of the optimization framework using simulated data follows the steps outlined below:

- Select a starting depth D_H representing the beginning of the lateral section of the target well.
- Select values for WOB and RPM, and use a physical model to compute the corresponding ROP over the first 100 ft of drilling, i.e., from D_H to $D_H + 100$ ft
- At depth $D_H + 100$ ft apply transfer learning to the pre-trained XGBoost ROP model using the simulated data from the first 100 feet.
- Between $D_H + 100$ and $D_H + 200$ ft, launch the optimization algorithm every 10 ft to identify drilling parameters that maximize the ROP.
- For each optimization point:
 - The algorithm uses the most recent updated ROP prediction model (after TL) to suggest a combination of drilling parameters.
 - These suggested parameters are then applied as if used in real drilling operations.
 - The actual ROP corresponding to these parameters is computed using the physical model.
- At $D_H + 200$ ft a second TL update is performed using all accumulated data from D_H to $D_H + 200$ ft.

Note: The predicted ROP (from the machine learning model) may differ from the actual ROP (computed via the physical model), just as in real operations where predictions are not always exact.

- Continue this cycle of TL every 100 ft and optimization every 10 ft as drilling progresses.

To evaluate the proposed framework, two simulated target wells are tested. Each well is generated using different physical model parameters and unique initialization values for WOB and RPM over the first 100 ft of drilling. Both test wells share the same starting depth of the lateral section, $D_H=12,200$ ft. As previously stated, the XGBoost model trained on real field drilling data is used as the source model for TL in both cases. During the optimization process, constraints are applied in the form of upper and lower bounds on surface rotation speed, downhole motor rotation speed, and WOB to ensure operational feasibility.

Fig. 7 and Fig. 8 illustrate the ROP heatmaps of the two simulated wells as a function of WOB and total RPM, generated using the physical model. In both figures, the red cross marks the initial drilling parameters at depth $D_H + 100$ ft (i.e., 12,300 ft), and the black cross indicates the drilling parameters suggested at the final depth (13,000 ft for the first test well and 13,700 ft for the second). The green line traces the sequence of drilling parameters proposed by the optimization algorithm every 10 ft throughout the lateral section. As shown, the optimization algorithm successfully converges toward the region of maximum ROP by iteratively suggesting drilling parameters every 10 ft. These suggested parameters progressively approach the optimal region, demonstrating the efficacy of the optimization cycle. This convergence is largely attributed to the use of transfer learning, which allows the model to gradually adapt to the specific characteristics of each test well. Since the simulated wells differ significantly from the real-world data used to train the original source model, TL plays a critical role in enabling accurate ROP prediction and effective parameter optimization.

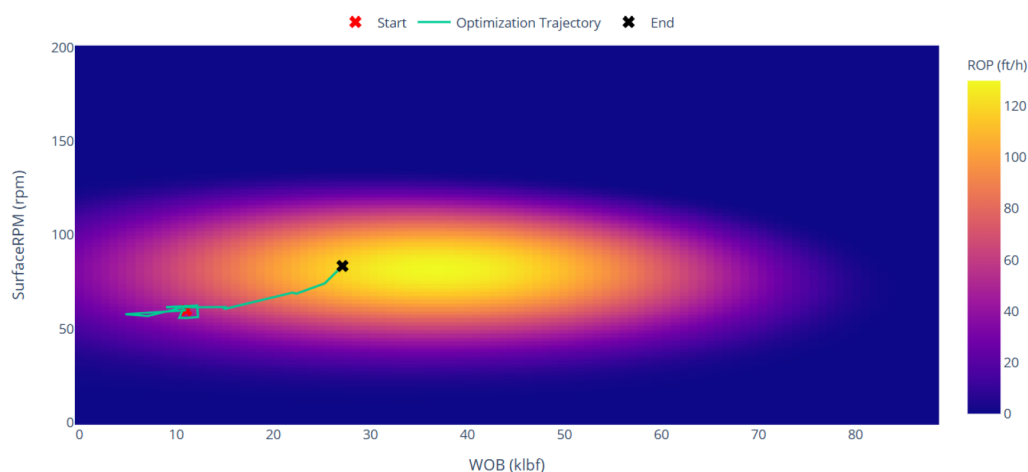


Figure 7—ROP heatmap of the first simulated test well as a function of WOB and total RPM, generated using the physical drilling model. The surface illustrates regions of high and low penetration rates. The red cross marks the starting point of the optimization algorithm, the black cross indicates the final point after convergence, and the green line traces the optimization path taken between them over 700 ft.

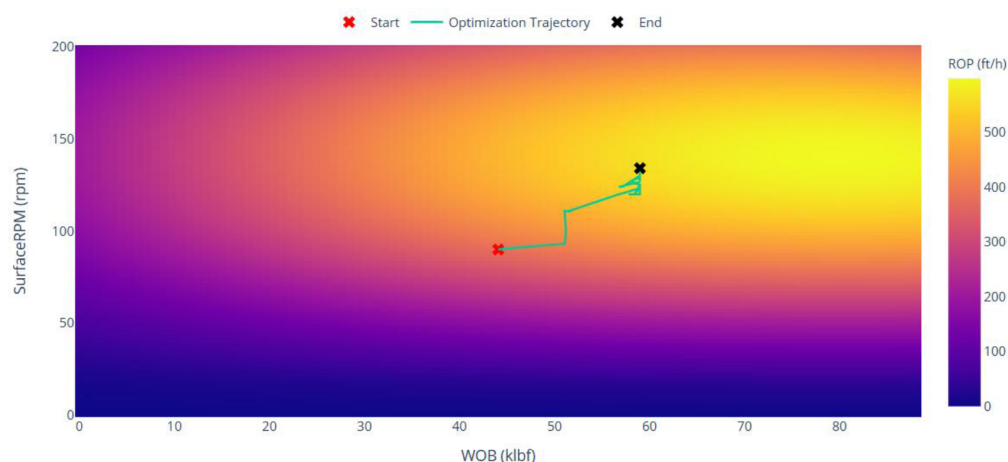


Figure 8—ROP heatmap of the second simulated test well as a function of WOB and total RPM, generated using the physical drilling model. The surface illustrates regions of high and low penetration rates. The red cross marks the starting point of the optimization algorithm, the black cross indicates the final point after convergence, and the green line traces the optimization path taken between them over 1400 ft.

Application on an actual rig

A horizontal well was drilled immediately after completing the lateral sections of wells from the same pad. The lateral section measured approximately 7,000 feet in length (Fig.9) and was drilled using a similar BHA as the previous lateral from the pad: an 8 $\frac{3}{4}$ " BHA with a mud motor. The featured ROP optimizer was used to enhance the drilling efficiency of this new lateral section.

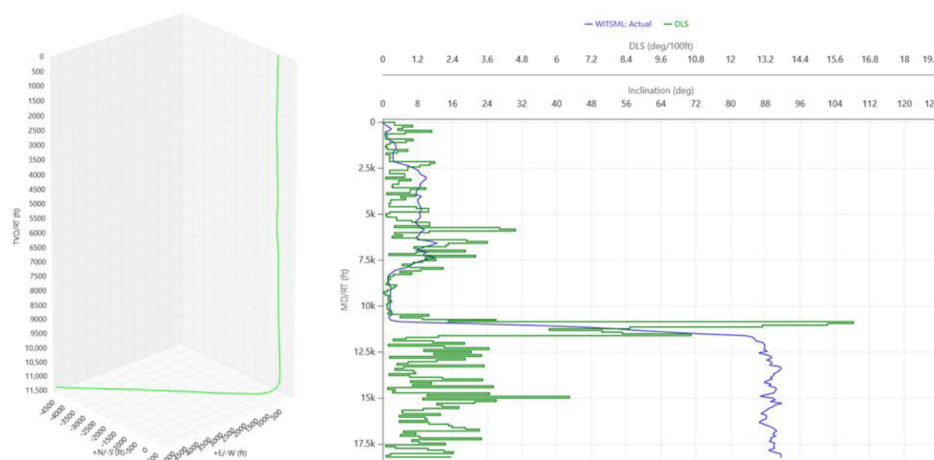


Figure 9—Wellbore trajectory of the unconventional well of this case history.

The process is as follows (Fig.10): The rig contractor's advanced engineering team and real-time operating center (ROC) collaborate with the operator to determine the offset wells to be considered. The ROC is tasked with setting up and monitoring the predictive vibration model and the ROP optimizer model. These models are fully integrated to the real-time drilling performance platform utilized by the team on a daily basis. Real-time WITSML data is received from the rig, and this data is processed by the ROP optimizer model to provide updated recommendations. The ROC also establishes the connection with the rig control system, allowing the autodriller setpoints— WOB, ROP, surface rotation speed, and differential pressure—to be regulated by the ROP optimizer model. With the driller's approval, updated recommendations are transmitted to the rig control system at regular drilled depth intervals. The updated recommendations are implemented without any further interaction from the driller at a regular interval.

The frequency and limits of these recommendations are agreed upon with the operator. While the driller maintains the ability to resume manual control whenever necessary, in this case, the ROP optimizer model governed operations for 97% of the rotate drilling once the test was started, during which the driller demonstrated 100% adherence to the model's recommendations.

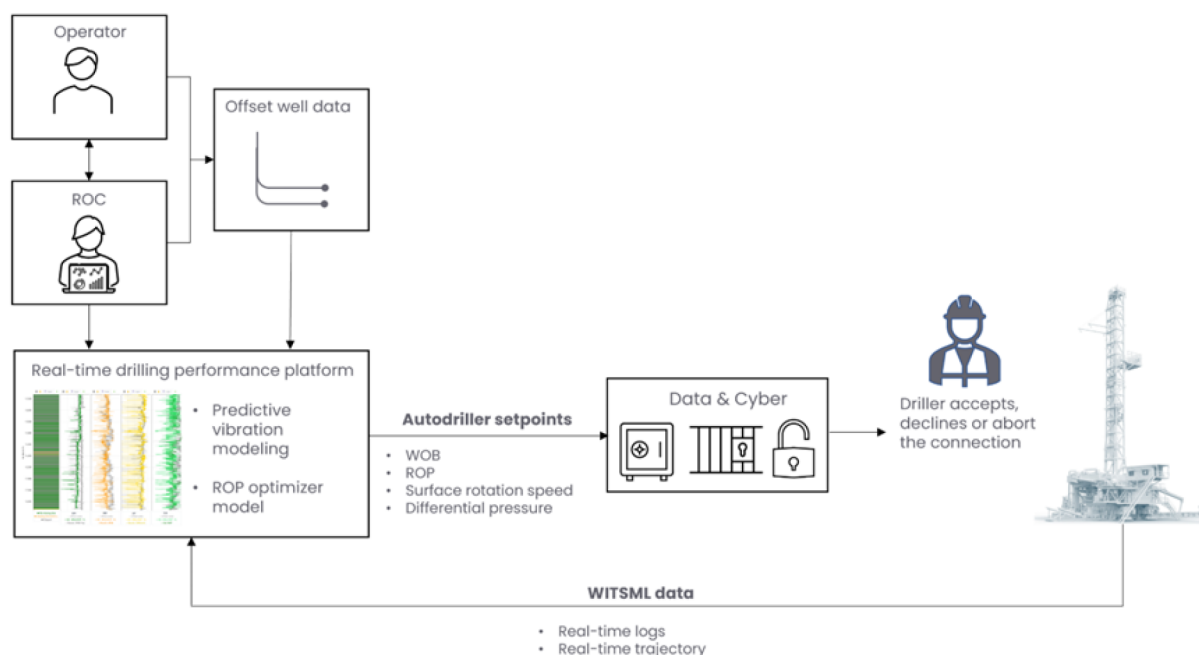


Figure 10—Real-time rig control ROP optimization process. The process includes collaboration between the well's operator and the rig contractor's real-time operating center and driller.

The strength of the featured solution lies in its seamless integration within the company's framework. Our approach benefits from the unified structure of having the ROC, drillers, and autodriller developers all under the same corporate umbrella. This cohesion ensures optimal communication and efficiency, resulting in superior operational performance and adherence.

In this case, two offset wells from the same pad, which demonstrated good overall performance, were selected. A real-time predictive vibration model was used to provide recommended surface rotation speeds to the ROP optimizer model to prevent harmful drilling vibrations (Fig.11). Additional constraints were included, such as operational limits on WOB and differential pressure, as defined by the driller. As soon as real-time data from the current well became available, they were prioritized with more weight than the offset well data in the ROP optimizer. As a result, the ROP optimizer model consistently delivered recommendations for WOB, ROP, differential pressure, and surface rotation speed. This led to an average ROP that was 26% higher compared to offsets. These offsets were drilled on the same pad, in the same formation, and with the same BHA design. Additionally, more stable drilling conditions were observed due to the optimizer making recommendations which performed well with the rigs control system. This additional stability allowed the parameter constraints from the operator to be more consistently utilized, which produced optimum performance without increasing the risk of failure.

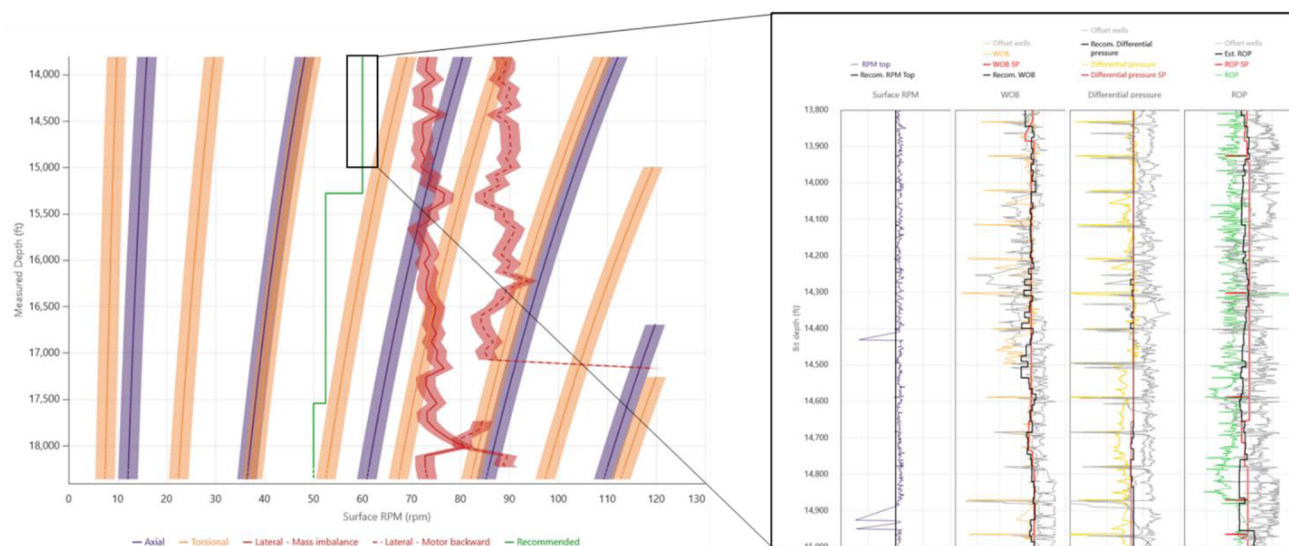


Figure 11—Left: Real-time predictive vibration model for the current well. Right: ROP optimizer results between 13800 ft and 15000 ft. The graph shows surface rotation speed, WOB, differential pressure and ROP actual values, recommendations generated by the ROP optimizer, autodriller setpoints and offset well data.

Conclusion

This work introduces a comprehensive, real-time ROP optimization framework that combines machine learning, transfer learning, and predictive vibration modeling within a constraint-aware optimization loop. By integrating data-driven predictions with physics-based constraints, the proposed method addresses key limitations of conventional ROP optimization approaches, namely, the lack of generalization across wells and the absence of operational safety considerations. The use of transfer learning allows the prediction model to continuously adapt to evolving drilling conditions, while the incorporation of vibration constraints ensures that optimized parameters remain within safe operational ranges. Simulation results show that the optimization algorithm consistently converges toward optimal drilling parameters, even under varying well conditions. Field validation confirms the applicability and robustness of the framework in real-world operations. Overall, this hybrid approach represents a significant advancement toward safer, smarter, and more efficient drilling. It provides drilling engineers with a practical tool for real-time decision-making that balances performance optimization with operational reliability.

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